# ELISA-constrained fit and its predictions · 10,000 simulated T cells · ΔFRO sensing · $R^2 = 0.94$ over the six measured profiles

**A** How the T cell profile is measured · 5,000 synthetic cells

**B** The fit: measured profiles and the fitted model

**C** The population over 72 h at the fitted tumor size, R = 1500 µm · rows 1–2 cells, rows 3–4 the same cells as density, rows 5–6 the same run at one particle per real T cell

**D** Four tumor sizes at 72 h · left half cells, right half the same cells as density · rows 3–4 repeat the density at one particle per real T cell

**E** Predicted T cell density at 72 h against distance, across those four sizes

synthetic cell
the line being sampled
the other rotations
1 · one line through the centre
2 · rotate it in 1° steps
3 · average every line profile
4 of the 180 single angles (clipped at the top)
the 180° average
T cell density (cells / mm²)
Distance from centre (mm)
measured mean (the fit target)
ELISA-constrained model
tumor edge, 1.5 mm
Off-target (parental) · 24 h
R² = 0.867
recruited 673 of 10,000
Off-target (parental) · 47 h
R² = 0.914
recruited 972 of 10,000
Off-target (parental) · 72 h
R² = 0.808
recruited 1,221 of 10,000
Target (CD19) · 24 h
R² = 0.821
recruited 849 of 10,000
Target (CD19) · 47 h
R² = 0.965
recruited 1,392 of 10,000
Target (CD19) · 72 h
R² = 0.929
recruited 1,738 of 10,000
Mean intensity (a.u.)
Distance from tumor centre (mm)
not yet recruited
recruited (same colour in both conditions)
tumor edge
T cell density (cells / mm²)
Cells Tumor only (no feedback)
0 h recruited 0
8 h recruited 352
16 h recruited 524
24 h recruited 662
32 h recruited 785
40 h recruited 885
48 h recruited 983
56 h recruited 1,056
64 h recruited 1,142
72 h recruited 1,223
1 mm
Cells Tumor + T cell feedback
0 h recruited 0
8 h recruited 361
16 h recruited 591
24 h recruited 840
32 h recruited 1,095
40 h recruited 1,285
48 h recruited 1,419
56 h recruited 1,532
64 h recruited 1,641
72 h recruited 1,755
Density Tumor only (no feedback)
Density Tumor + T cell feedback
Density, 200k check Tumor only (no feedback)
Density, 200k check Tumor + T cell feedback
0 h
8 h
16 h
24 h
32 h
40 h
48 h
56 h
64 h
72 h
Tumor only (no feedback)
R = 750 µm cells · recruited 837
R = 1500 µm cells · recruited 1,223
R = 2250 µm cells · recruited 1,638
R = 3500 µm cells · recruited 2,319
R = 750 µm density
R = 1500 µm density
R = 2250 µm density
R = 3500 µm density
Tumor + T cell feedback
R = 750 µm cells · recruited 1,243
R = 1500 µm cells · recruited 1,755
R = 2250 µm cells · recruited 2,278
R = 3500 µm cells · recruited 3,243
R = 750 µm density, 200k particles
R = 1500 µm density, 200k particles
R = 2250 µm density, 200k particles
R = 3500 µm density, 200k particles
Tumor only
Tumor + T cell positive feedback
tumor edge
Tumor radius 750 µm
Tumor radius 1500 µm (the fitted size)
Tumor radius 2250 µm
Tumor radius 3500 µm
T cell density (cells / mm²)
Distance from tumor centre (mm)